ELEC5506: Process instrumentation & Control

Dual-conveyor workcell system

Objective

The dual-conveyor workcell system consists of,

- Two conveyors with sensors and actuators
- Micro logix 1400 series PLC, and
- Computer with RS Logix software to programmatically control the system.

For this lab, you will be working with both conveyors that will be used to sort washers based on their size (height) and composition material (plastic or metal).

General overview of the top conveyor belt

The following components are used to carry out the identification and sorting of the washers:

- The top conveyor is used to carry the washers.
- there are three sizes of washer: too big, too small and just right.
- One dispenser is used to dispense (drop) the washers on the conveyor Opto sensors are used to sense the presence of washers A height sensor is used to measure the height of the washers.
- An inductive detector identifies the metal parts so that the metal washers can be separated from plastic washers
- Solenoid operated flippers are used to ”flip” the washers into appropriate chutes. Only washers with correct height will be flipped into the chutes. Washers with incorrect height will be sent to the bin. Metal washers with correct height will be flipped to Chute 1 whereas plastic washers with correct height will be flipped to Chute 2.

For detailed information on the operation of the whole system for the top conveyor, please watch the video uploaded in LMS.

Exercises

Please remember to come into your lab session prepared, you can use the emulator to check your ladder logic before attending your lab session. Also remember to:-

- Watch the video to understand the sequence and the usage of sensors and actuators to control the sequence. There is a video uploaded in same the section as the recorded lectures entitled “Dual-conveyor workcell system” which covers the details required to complete the lab.

- Use the template provided to you in LMS. Download the template and save it to your PC (Right-click>Save link as), then open it using RSLogix. Only groups who send their ladder logic using the templates will get their codes tested. There is a video

uploaded in same the section as the recorded lectures entitled “Emulating with RSLogix” which goes over what needs to be done.

- The inputs and output bit addresses are provided in LMS as well.
- You can simulate the event of a washer arriving at the sensor by manually toggling the bit in the Data Files for the inputs (ie: OPT1, OPT2, etc)

Exercise 1:
Write a program to dispense washers from dispenser 1 and measure their height, discarding incorrect height washers over the back end. Use a timer to stop the washer when it comes directly under the height sensor. Consider how you will control the release of each washer.

Exercise 2.
Extending the above program, identify whether washers of correct height are metal or plastic and actuate the flippers to separate metal and plastic washers (i.e. metal washers must be pushed into one Chute 1 and plastic washers must be forced into Chute 2).

Exercise 3
Dispense the washers from Chutes 1 and 2 and sort using Sol 4 into separate streams. For full marks this should run concurrent with the sorting on conveyor 1.

Each exercise must:

- Be shown to the facilitator separately before attempting the next
- Be able to distinguish between all sizes and types of washer
- Only run the conveyor belts when a washer is on it.

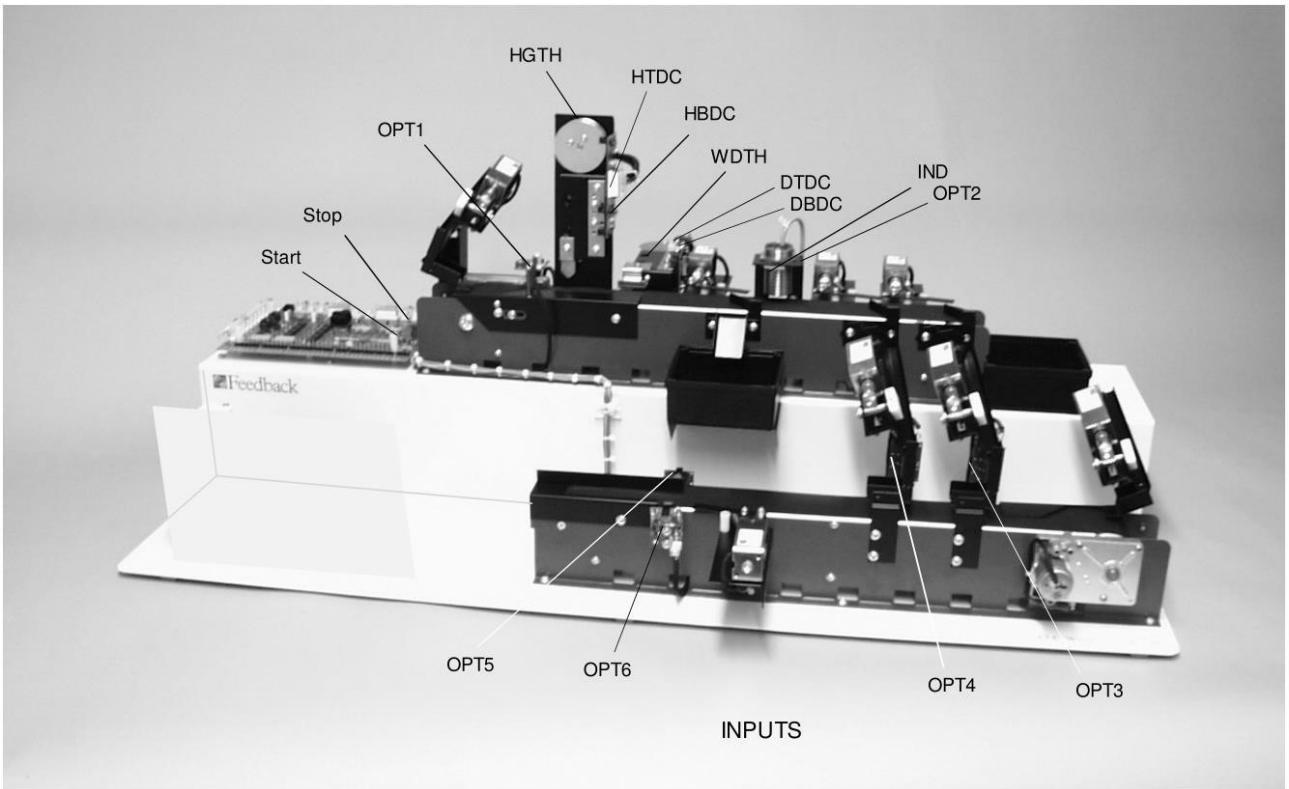

Figure 1: Inputs of the dual-conveyor workcell system (WDTH, DTDC & DBDC is not present in our system)

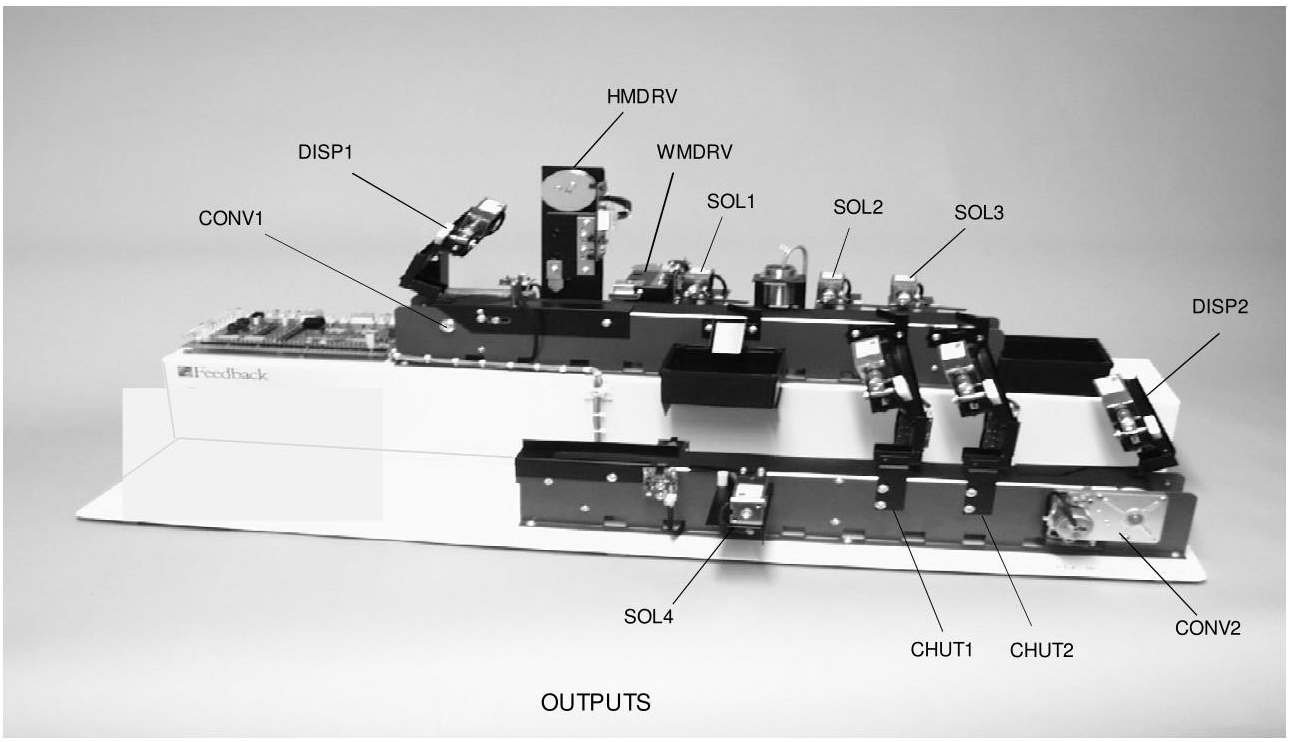

Figure 2: Outputs of the dual-conveyor workcell system (WMDRV is not present in our system)